ECEN 5692: HW3 | Matt Hartnett

Problem 1.2.A.11

I have accidentally already solved this as I thought it was due as part of HW2. See the underlined problem on the next page for the solution.

Problem 1.3.A.1

$$f_{\hat{X}}(\hat{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma_{\hat{X}})}} e^{-\frac{1}{2}(\hat{x}-\vec{\mu}_{\hat{X}})^T \Sigma_{\hat{X}}^{-1} (\hat{x}-\vec{\mu}_{\hat{X}})}$$

$$\Sigma_{\hat{X}} = \begin{bmatrix} \sigma_{x_1}^2, & Cov(X_1, X_2) \\ Cov(X_2, X_1), & \sigma_{x_2}^2 \end{bmatrix}$$

$$Cov(X_2, X_1) = E[X_1 X_2] - \mu_1 \mu_2 = Cov(X_1, X_2)$$
$$= -2 - (-3)(2)$$
$$= 4$$

$$\Sigma_{\hat{X}} = \begin{bmatrix} 9 & 4 \\ 4 & 7 \end{bmatrix} \rightarrow \text{Has to be A or C}$$

$$\vec{\mu}_{\hat{X}} = \begin{bmatrix} \mu_{x1} \\ \mu_{x2} \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

We can then fill in the full formula above to see the answer is $\boxed{C}$

Problem 1.3.A.2

1) $$\Sigma_{\hat{Z}} = \begin{bmatrix} \sigma_{z_1}^2, & Cov(Z_1, Z_2) \\ Cov(Z_2, Z_1) & \sigma_{z_2}^2 \end{bmatrix}$$

$$Cov(X_1, X_2) = E[X_1 X_2^*] - \mu_1 \mu_2^*$$

$$Cov(Z_1, Z_2) = (4-j) - (-1)(2-j)$$
$$= 6-2j$$

$$Cov(Z_2, Z_1) = E[Z_2 Z_1^*] - \mu_{z_2} \mu_{z_1}^*$$

$$E[Z_2 Z_1^*] = E[Z_1 Z_2^*]^* \quad \text{since } E \text{ is linear}$$
$$= (4+j)$$

$$Cov(Z_2, Z_1) = (4+j) - (2+j)(-1)$$
$$= 6+2j$$

$$\Sigma_{\hat{Z}} = \begin{bmatrix} 16, & 6-2j \\ 6+2j, & 25 \end{bmatrix} = \boxed{B}$$

Note: continued on pg 3 due to 1.2.A.11

Problem 1.2.A.10 (cont.):

$$y(t) = 2(e^{j2\pi f_c t} e^{-j(2\pi f_c t + \theta')}) * h_L(t)$$

$$= 2e^{-j\theta'} * h_L(t)$$

$$\boxed{y(t) = 2e^{-j\theta'}}$$

Problem 1.2.A.11:

$$x(t) = 1$$

$$x_c(t) = R_e[x(t)\sqrt{2}e^{j2\pi f_c t}]$$
$$= \sqrt{2}e^{j2\pi f_c t}$$

$$y(t) = (x_c(t)\sqrt{2}e^{-j2\pi f_c' t}) * h_L(t)$$

$$= 2(e^{j2\pi f_c t} e^{-j2\pi f_c' t}) * h_L(t)$$

$$= 2e^{j2\pi(f_c - f_c')t} * h_L(t)$$

If we assume that the frequency difference is less than the LPF's cut off frequency:

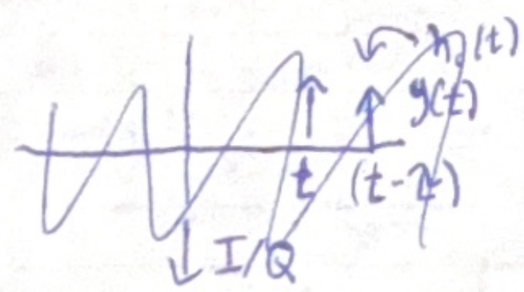
$$\boxed{y(t) = 2e^{j2\pi(f_c - f_c')t}}$$

If $(f_c - f_c') > \omega_{toff}$, $y(t) = \emptyset$

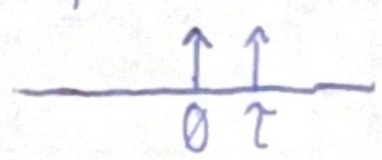
Problem 1.2.A.12:

$$h_c(t) = \delta(t) + a\delta(t - \tau_1)$$

1. $h(t) = \frac{1}{\sqrt{2}}$ I/Q↓ $(h_c(t))$



Graph of $h_c(t)$:



$$\delta(t-\tau)g(t) = \delta(\tau)g(\tau)$$

I/Q↓: $\sqrt{2}\, h_c(t) e^{-j2\pi f_c t} * h_L(t)$

$$h(t) = (\delta(t) + a\delta(t-\tau))e^{-j2\pi f_c t} * h_L(t)$$

$$\cancel{= (1 + ae^{-j2\pi f_c \tau}) * h_L(t)}$$

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Problem 1.3.A.2 (cont.)

2) From looking at the possible options and the answer to part 1, we can see the answer is $\boxed{C}$ since it is the only one with the correct covariance $\Sigma_{\vec{z}}$.

3) $\vec{z} = \begin{bmatrix} -1 \\ 2+j \end{bmatrix} \Rightarrow \vec{z}_I = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$E[z_{1I} z_{1I}^*] = -1$

$Cov(z_1, z_2) = -1 - (0)(1)$
$= -1$

Not an option, this is wrong ↑

$\Sigma_{\vec{z}_I} = E\left[\begin{bmatrix} Im(z_1 - \mu_{z_1}) \\ Im(z_2 - \mu_{z_2}) \end{bmatrix} [Im(z_1 - \mu_{z_1}) \;\; Im(z_2 - \mu_{z_2})]\right]$

$= E\left[\left[Im(-1 -\right.\right.$

$\Sigma_{\vec{z}_I} = \frac{1}{2} Re[\Sigma_{\vec{z}_1}]$

$= \frac{1}{2}\begin{bmatrix} 16 & 6 \\ 6 & 25 \end{bmatrix}$

$= \begin{bmatrix} 8 & 3 \\ 3 & 12.5 \end{bmatrix}$ $\boxed{A}$

4) For CSCG $\Sigma_{\vec{z}Q} = \Sigma_{\vec{z}I} \Rightarrow \boxed{D}$

5) $\Sigma_{\vec{z}_I \vec{z}_Q} = -\frac{1}{2} Im[\Sigma_{\vec{z}}]$

$= -\frac{1}{2}\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \Rightarrow \boxed{C}$

6) $\Sigma_{\vec{z}_Q \vec{z}_I} = \Sigma_{\vec{z}_{IQ}} \Rightarrow \boxed{C}$

7) $\vec{z} = \begin{bmatrix} -1 \\ 2+j \end{bmatrix}$, $z_1 = -1$, $z_{1I} = -1$

$\Sigma_{\vec{z}_I} = \frac{1}{2}\begin{bmatrix} Re[\Sigma_{\vec{z}}] & -Im[\Sigma_{\vec{z}}] \\ Im[\Sigma_{\vec{z}}] & Re[\Sigma_{\vec{z}}] \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \sigma^2 & Cov(z_1, z_2) \\ Cov(z_2, z_1) & \sigma^2 \end{bmatrix}$

$\Sigma_{\vec{z}_I} = \frac{1}{2}\sigma^2 = \frac{1}{2}(16) = 8$

$f_{z_{1I}}(z_{1I}) = \frac{1}{\sqrt{\pi 2 \cdot 8}} e^{-\frac{(z_{1I}+1)^2}{2 \cdot 8}} \Rightarrow \boxed{B}$

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Problem 1.3.A.3:

$\bar{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = 0$

$E[z_1 z_2^*] = 2-4j$
$\sigma_{z1}^2 = 48, \quad \sigma_{z2}^2 = 30$

1) $\Sigma_z = \begin{bmatrix} \sigma_{z_1}^2 & Cov(z_1, z_2) \\ Cov(z_2, z_1) & \sigma_{z_2}^2 \end{bmatrix}$

$Cov(z_1, z_2) = \cancel{Cov(z_2, z_1)} = E[z_1 z_2^*] - \mu_1 \mu_2 = 2-4j$ $\quad Cov(z_2, z_1) = E[z_1 z_2^*]^* - \mu_2 \mu_1 = 2+4j$

$\Sigma_z = \begin{bmatrix} 48 & 2-4j \\ 2+4j & 30 \end{bmatrix} \Rightarrow \boxed{B}$

2) $\Sigma_{\hat{z}} = \begin{bmatrix} \Sigma_{\bar{z}_I} & \Sigma_{\bar{z}_I \bar{z}_Q} \\ \Sigma_{\bar{z}_I \bar{z}_Q} & \Sigma_{\bar{z}_Q} \end{bmatrix}$

$\Sigma_{\bar{z}_I} = \frac{1}{2} Re[\Sigma_z] = \Sigma_{\bar{z}_Q}$

$= \begin{bmatrix} 24 & 1 \\ 1 & 15 \end{bmatrix}$

$\Sigma_{\bar{z}_I \bar{z}_Q} = -\frac{1}{2} Im[\Sigma_z]$

$= \begin{bmatrix} 0 & +2 \\ -2 & 0 \end{bmatrix}$

$\Sigma_{\hat{z}} = \begin{bmatrix} 24 & 1 & 0 & +2 \\ 1 & 15 & -2 & 0 \\ 0 & +2 & 24 & 1 \\ -2 & 0 & 1 & 15 \end{bmatrix} \Rightarrow \boxed{C}$